

# SHUVANGEE SETH

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Computer Science senior at Illinois Tech (GPA 3.7) with hands-on experience in ML engineering, full-stack development, and teaching. Built and deployed an ML pipeline processing 10K+ data points at Aether Intelligence. Passionate about scalable systems, data-driven solutions, and making tech accessible.

## EDUCATION

**Illinois Institute of Technology**, Chicago, IL

Aug 2023 – May 2027

*Bachelor of Science in Computer Science* | **GPA: 3.7 / 4.0** | Dean's List — 4 consecutive semesters (Spring '24 – Spring'26)

**Coursework:** Data Structures & Algorithms, System Programming, Assembly Language, Discrete Structures, Database Organization

## TECHNICAL SKILLS

**Languages:** Python, Java, C/C++, JavaScript, Go, Ruby, SQL, HTML/CSS

**Tools & Platforms:** MySQL, Git/GitHub, Linux/Bash, REST APIs, Tableau, Excel, A/B Testing, Agile/Scrum

**Libraries & Frameworks:** Scikit-learn, Pandas, NumPy, MySQL Connector, Tabulate, NASA NeoWs API

## EXPERIENCE

**Illinois Institute of Technology** | *Teaching Assistant – Systems Programming (CS 351)*

Jan 2026 – Present

- Facilitate weekly lab sessions for **40+ students** on C programming, memory management, and POSIX system calls
- Debug and review student code submissions, delivering targeted feedback that improved average assignment scores by **12%**
- Develop supplementary lab exercises on pointer arithmetic and process scheduling, adopted by the course for **3 sections**

**Aether Intelligence** | *Software Engineering Intern*

Aug 2024 – Dec 2024

- Engineered a crop recommendation ML pipeline in **Python + MySQL**, ingesting and transforming **10,000+ weather data points**
- Boosted prediction accuracy from **74% to 87%** by benchmarking 5 classification models and applying feature engineering with cross-validation
- Deployed the optimized model as a **RESTful API endpoint**, enabling real-time agricultural predictions for downstream consumers

**Microsoft TEALS** | *Teaching Assistant*

Aug 2024 – Present

- Lead weekly **3-hour interactive labs** for **30 high school students**, delivering College Board's AP CS Principles curriculum
- Designed hands-on coding exercises in Scratch and Python, increasing student engagement and **boosting exam pass rates by 20%**

**Illinois Institute of Technology** | *Teaching Assistant – DSA & Intro to CS*

Aug 2024 – Dec 2025

- Conducted lab sessions and office hours for **100+ students** across 2 courses, reinforcing recursion, sorting, trees, and OOP
- Graded **200+ assignments/semester**, identified recurring error patterns, and collaborated with faculty to refine **4 lab modules**

## PROJECTS

**Research Compass AI** | Next.js, FastAPI, PostgreSQL, OpenAI, AWS, Neo4j, Vector Database

June 2026

- Engineered an AI-powered research exploration platform that analyzes academic literature, maps citation networks, and identifies promising research directions for early-stage researchers.
- Processed and indexed **10,000+** research papers using Semantic Scholar, arXiv, and OpenAlex APIs, building a searchable knowledge base enriched with embeddings and metadata.
- Designed a research-gap detection engine leveraging RAG, graph analytics, topic clustering, and LLM-based reasoning to uncover underexplored areas and generate project recommendations.
- Deployed a full-stack cloud solution on AWS with automated data pipelines, semantic search, personalized dashboards, and AI-generated literature reviews.

**Crop Recommendation ML Pipeline** | Python, Scikit-learn, MySQL, REST API

Aug – Dec 2024

- Built an end-to-end pipeline: data ingestion → cleaning → feature engineering → model training → API deployment
- Achieved **87% accuracy** by comparing Random Forest, KNN, SVM, Naïve Bayes, and Decision Tree classifiers

**F1 Database Management System** | Python, MySQL, Tabulate

Aug – Dec 2024

- Architected a role-based CLI with **Admin** (CRUD) and **User** (read-only) modes over a normalized schema of **8 relational tables**
- Optimized complex JOIN queries with indexing and restructuring, reducing average response time by **40%**

**Data Visualization Portfolio** | Tableau, Excel, A/B Testing

Jan – May 2025

- Analyzed **5 real-world datasets**, identified KPIs, ran A/B tests, and built interactive Tableau dashboards with drill-down filters

## LEADERSHIP & ACTIVITIES

**President, ACM-W (Women in Computing)** – Illinois Tech

Jan 2025 – Present

- Spearhead diversity-in-tech initiatives, organizing **8+ events/semester** and mentoring **15+ underrepresented CS students**

**SOAR Orientation Leader** – Illinois Tech

Jun 2025 – Present

- Guide **900+ incoming freshmen and graduate students** through orientation; serve as primary contact via phone, email, and social platforms

**Accessibility Hackathon** – Google Chicago

Apr 2025

- Built an accessibility-focused **Python prototype** in a cross-functional team; presented to a panel of Google engineers and won recognition

**President, F1 Club** – Illinois Tech

Jan 2025 – Present

- Coordinate club events and community engagement for **50+ members**, managing budgets and sponsorship outreach

**Also:** GHC '25 Attendee | Community Desk Assistant, IIT Residence Life (Oct 2024–Present) | P33 Chicago Strong Start | Cognizant Agile Certification